

# Jiarui (Ray) Xu

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## Summary

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**Software Engineer** with **5+ years** of experience delivering production software across **backend services, web applications, and system integration/testing** at **Pax Technology, Bank of America/Shift4** ecosystems, **Oracle-based enterprise systems**, and academic research. Strong in **Python, Java, JavaScript, SQL, REST APIs**, and **object-oriented design**. Proven track record in **debugging, automated testing, CI/CD**, and building reliable systems with real-world users at scale. **AWS Certified Solutions Architect – Associate. ISTQB CTFL.**

## Skills

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**Languages:** Python, Java, JavaScript, SQL, C#, C++

**Backend / APIs:** RESTful services, request/response modeling, auth/session basics, error handling, retries/timeouts, logging/monitoring

**Databases:** MySQL, SQLite, Oracle; schema design, indexing basics, query optimization, data validation

**Web / Frontend:** JavaScript, Vue.js, React (project experience), HTML/CSS

**Cloud / DevOps:** AWS (EC2, S3, Lambda, IAM, CloudWatch), Docker, CI/CD (Jenkins, GitHub)

**Testing:** JUnit, PyTest, Postman, Selenium, Appium; regression suites, test automation, defect triage

**Tools:** Git, Gradle, IntelliJ, Android Studio, Linux, SonarQube, Jira

## Experience

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### SIT Engineer

Pax Technology Inc. — Jacksonville, FL (2024–Present)

- Delivered production features for high-availability payment platforms by implementing and integrating **transaction workflows** (online/offline, EMV/contactless) with robust **error handling, timeouts, and recovery** logic.
- Wrote and maintained **production Java modules** across configuration management, procedure/action orchestration, transmission, receipt rendering, and data providers; improved maintainability via clearer interfaces and contract-driven data models.
- Built **host integration layers** (RPC/HTTP-style payloads) including request/response mapping, protocol compliance checks, and defensive parsing; partnered with cross-functional teams to resolve integration issues and ship fixes.
- Developed and maintained **merchant-facing web apps** (Vue.js + Cordova/WebView) supporting multi-step flows, device communication (USB/Serial/Bluetooth), offline behavior, and resilient UI state management.
- Performed deep **debugging and troubleshooting** using logs, stack traces, and lifecycle analysis; owned issue triage and root-cause analysis and collaborated on fixes across teams.
- Built automated testing frameworks in **Python** (PyTest + Appium) for functional/UI/regression testing; created API validation scripts (Postman + Python) and tooling for log parsing and stress/regression runs.
- Contributed to **CI/CD pipelines** (Jenkins + Gradle + Git) enabling multi-flavor builds, automated versioning, release validation, and regression gates.
- Technologies: **Java, Python, JavaScript**, Vue.js, Docker, Jenkins, Gradle, Git, SonarQube, SQLite/MySQL, REST/RPC.

### Android Developer

BeaconFire Inc. — Plainsboro, NJ (2023–2024)

- Built user-facing features with **Kotlin** using clean separation of concerns (MVVM), emphasizing maintainable architecture and testability.
- Integrated **REST APIs** using Retrofit/OkHttp; implemented request models, error handling, and client-side caching patterns to improve reliability.
- Strengthened quality via **debugging**, logging, and automated tests (Espresso/Jacoco), improving defect turnaround and release confidence.

### Software Engineer

Stellar Services — Atlanta, GA (2023)

- Developed enterprise web features integrating **Oracle Database** with **ASP.NET (C#)**; implemented server-side logic and data access patterns.
- Built JavaScript-based UI components and delivered end-to-end features including data retrieval, validation, and document generation workflows.

#### Artificial Intelligence Teaching Assistant Georgia Institute of Technology — Atlanta, GA (2022–2023)

- Supported graduate and undergraduate courses through office hours and project guidance; helped students debug, optimize, and reason about algorithms and systems.
- Built research prototypes in **Python/PyTorch** (multi-task NLP, segmentation, and audio transcription experiments), emphasizing reproducible pipelines and evaluation.

#### Software Engineer Zhejiang University — Hangzhou, China (2020–2021)

- Built a **Python/C++** music analysis and generation system (tonality recognition, chord modeling, melody generation) with production-style data processing pipelines.
- Delivered a **full-stack web platform** enabling users to generate background music for uploaded videos; supported real users during official events (**5,000+ users**).
- Website: Web Platform

## Projects

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#### JIUZIHUN (Nine Souls) — Online Multiplayer Strategy Card Game Independent Full-Stack Project

- Built a browser-based game with **JavaScript** and a backend server + database deployed on **Render**; implemented matchmaking, state synchronization, and persistence.
- Designed data-driven game logic and an AI decision engine; engineered reusable models for skills/cards/actions with clear interfaces and deterministic evaluation.
- Live Demo: Playable Web Version    Code: Frontend — Backend

#### Twitch Plus — Resource Recommendation Engine BeaconFire Inc.

- Developed a **Spring Boot** backend with RESTful APIs; integrated external APIs, stored data in **MySQL (AWS RDS)**, and implemented caching for performance.

#### US BroadPOS Automated Testing Framework (Python, UI Automation) Pax Technology

- Built automated regression suites in **Python/PyTest** with device orchestration, APK deployment, log collection, and HTML reporting to support reliable releases.

## Certifications

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- AWS Certified Solutions Architect — Associate
- ISTQB Foundation (CTFL)

## Education

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#### Master of Science in Computer Science Georgia Institute of Technology (2022–2023)

- GPA: 4.0/4.0    *Coursework:* Operating Systems, Data Structures, Algorithms, Distributed Systems

#### Master of Science in Music Technology Georgia Institute of Technology (2021–2023)

- GPA: 3.8/4.0

#### Bachelor of Electronic Engineering Xi'an Jiaotong University (2016–2020)

- GPA: 3.62/4.0